



ecoinvent 3.11 Dataset Documentation

'washing machine production – GLO – washing machine'

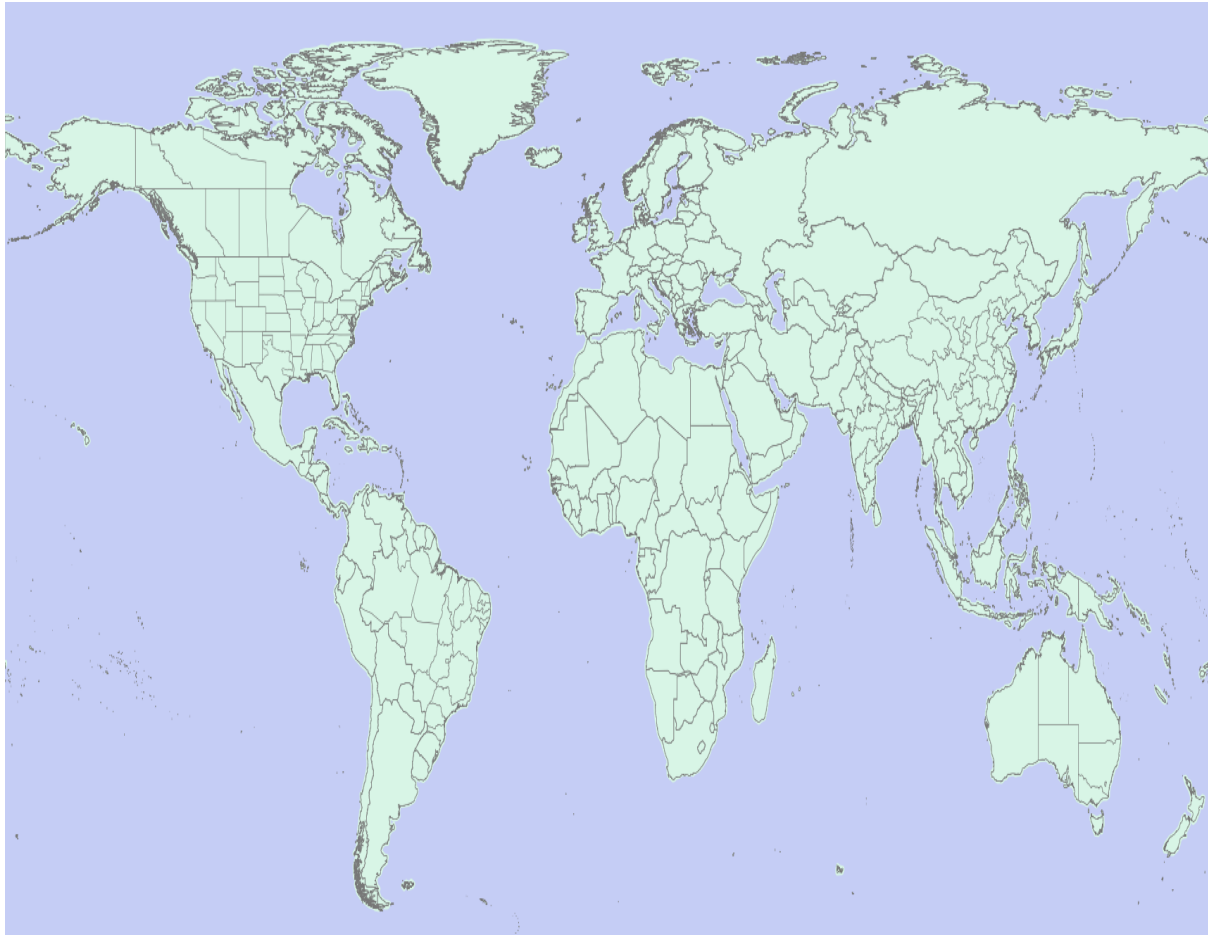
Note: This document contains only an extract of the information in the dataset. Additional data about properties of exchanges, mathematical relations, parameters, and contact information for authors and reviewers are available within the dataset, i.e. in ecoSpold format.

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Dataset Identification

Activity name	washing machine production
Geography	Global
Time period	2004-01-01 to 2024-12-31 - Valid for the entire period
Synonyms	laundry appliance, top-loaded washing machine, washer
ISIC rev.4 ecoinvent	2750:Manufacture of domestic appliances
Reference product	washing machine
CPC classification	44812: Dishwashing machines and clothes or linen washing or drying machines, household type, electric or non-electric
Dataset type	Ordinary transforming activity
Technology level	New
Version - system model	3.11 - Allocation, cut-off



Water Geography

Dataset Authorship

Data entry	Viachaslau Filimonau,
Data generator	Viachaslau Filimonau,

Exchange Summary

Reference product	Byproduct classification	Amount
washing machine	allocatable product	1 unit
Inputs from technosphere		Amount
acrylonitrile-butadiene-styrene copolymer		21.2 kg

aluminium, cast alloy	5.76 kg
cable, unspecified	0.63 kg
cast iron	0.92 kg
compressed air, 1000 kPa gauge	3.7 m3
concrete block	0.2 kg
copper, cathode	1.81 kg
electricity, high voltage	1.34e+2 kWh
energy and auxilliary inputs, metal working factory	3.43e-08 kg
epoxy resin, liquid	0.167 kg
flat glass, uncoated	1.8 kg
heat, district or industrial, other than natural gas	19.3 MJ
kraft paper	1.3 kg
metal working factory	3.43e-08 unit
nylon 6	0.1 kg
polyethylene, low density, granulate	0.2 kg
polypropylene, granulate	7.47 kg
polystyrene, general purpose	0.5 kg
polyvinyl chloride, unspecified polymerisation, weighted average	1.03 kg
printed paper	0.15 kg
seal, natural rubber based	0.01 kg
steam, in chemical industry	1.37 kg
steel, low-alloyed, hot rolled	31.1 kg
synthetic rubber	0.88 kg
tap water	11.5 kg
trisodium phosphate	0.52 kg

Inputs from technosphere, wastes	Amount
inert waste	-0.2 kg
used cable	-0.63 kg
waste aluminium	-5.76 kg
waste copper	-1.81 kg
waste glass	-1.8 kg
waste plastic, mixture	-21.2 kg

waste polyethylene	-0.3 kg
waste polypropylene	-7.47 kg
waste polystyrene	-0.5 kg
waste polyvinylchloride	-1.2 kg
waste rubber, unspecified	-0.89 kg
waste steel	-31.1 kg
wastewater, average	-0.0115 m3

Inputs from technosphere, recyclable materials	Amount
iron scrap, unsorted	-0.92 kg
waste packaging paper	-1.45 kg

Dataset Description

General comment

This activity represents the production and disposal of an average washing machine which is 75 kg in mass. It has the envisaged lifetime of 10 years and is manufactured in China. The figures on material inputs from technosphere and electricity required to produce an average washing machine are taken from Yuan et al. 2016. The figure on the annual production volume is estimated via GDP scaling. This is based on the annual amount of washing machine sales in EU as taken from Michel et al. 2015.

Included activities start

This activity starts with the reception of materials at the factory.

Included activities end

The activity includes the processes of material and energy use in manufacturing an average washing machine. The inventory additionally accounts for disposal of all materials. Packaging is not included.

Sampling procedure

Manufacturing data are taken from Yuan et al. 2016. Data on the annual production volume are taken from Michel et al. 2015. Sources: Yuan, Z., Zhang, Y., and Liu, X., 2016. Life cycle assessment of horizontal-axis washing machines in China. International Journal of Life Cycle Assessment, 21, 15-28. Michel, A., Josephy, B., Bush, E., et al. 2015. Monitoring the washing machines market in Europe. Discussion paper. Available from: http://www.topten.eu/uploads/File/EEDAL15_Annette_Michel_Monitoring_washing_machines_market.pdf (Accessed 20 January 2018).

Extrapolations

The figure on the annual world production volume is extrapolated from the annual volume of washing machines sold in EU. According to Michel, A., Josephy, B., Bush, E., et al. 2015. Monitoring the washing machines market in Europe. Discussion paper. Available from: http://www.topten.eu/uploads/File/EEDAL15_Annette_Michel_Monitoring_washing_machines_market.pdf (Accessed 20 January 2018), the annual sales of washing machines in EU in 2014 were equal to 15.2 million units. Then gross domestic product (GDP) at current prices (or GDP at prices of the current reporting period), also known as nominal GDP, was used. The EU contribution to global GDP in 2014 is 23.8% (Source: Eurostat 2016. The EU in the world - economy and finance. Available from: http://ec.europa.eu/eurostat/statistics-explained/index.php/The_EU_in_the_world_-_economy_and_finance (Accessed 14 January 2018). Consequently, the global production volume of washing machines can be estimated as $15.2 \text{ million} \times 100 / 23.8 = 63.87 \text{ million units}$.

Technology comment

The literature sources used refer to manufacturing in China.

Geography comment

Geography GLO was chosen because the dataset aims to represent the production of an average washing machine. The literature sources used refer to manufacturing in China.

Time period comment

The literature sources used refer to primary data from 2004. Although the background study utilised in this project (Yuan et al. 2016) was published in 2016, it refers to technology described as being 'current'.

Detailed Information For Exchanges

Reference product	Annual prod.vol.	Amount
washing machine	6.39e+7 unit	1 unit

Production volume: 6.39e+7 unit

Production volume comment: The figure on the annual world production volume is extrapolated from the annual volume of washing machines sold in EU as recorded by Michel et al. 2015.

Source: Michel, A. (2015)

Inputs from technosphere	Amount
acrylonitrile-butadiene-styrene copolymer	21.2 kg
Comment: Data from Yuan et al. 2016. In terms of outputs, assumed to have gone to 'waste plastic, mixture'.	
Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
aluminium, cast alloy	5.76 kg
Comment: Data from Yuan et al. 2016.	
Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
cable, unspecified	0.63 kg
Comment: This product represents average, unspecified cable. Data from Yuan et al. 2016.	
Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
cast iron	0.92 kg
Comment: Used here in place of iron. Data from Yuan et al. 2016.	
Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
compressed air, 1000 kPa gauge	1.07 m3
Comment: Data from Yuan et al. 2016.	
Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
compressed air, 1000 kPa gauge	2.63 m3
Comment: Data from Yuan et al. 2016.	
Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
concrete block	0.00032 kg
Comment: Data from Yuan et al. 2016. In terms of outputs assumed to have gone to inert waste.	
Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
concrete block	0.00154 kg
Comment: Data from Yuan et al. 2016. In terms of outputs assumed to have gone to inert waste.	
Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
concrete block	0.0034 kg
Comment: Data from Yuan et al. 2016. In terms of outputs assumed to have gone to inert waste.	
Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
concrete block	0.195 kg
Comment: Data from Yuan et al. 2016. In terms of outputs assumed to have gone to inert waste.	
Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	

copper, cathode	1.81 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
electricity, high voltage	1.34e+2 kWh
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
energy and auxilliary inputs, metal working factory	2.81e-08 kg
Comment: Estimated value. The input of infrastructure is approximated with the exchange 'metal working factory', this was chosen as metal is the main material constituting the final product. The general comment for the activity 'metal working faculty, GLO' [accessed 19.01.2018] specifies: '4.58E-10 units of this process are needed to produce 1 kg of final product. Additionally, the dataset for 'metal working factory' needs to be used together with the 'energy and auxilliary inputs, metal working factory', 1 kg of this exchange is required per 1 kg of final product. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
energy and auxilliary inputs, metal working factory	6.26e-09 kg
Comment: Estimated value. The input of infrastructure is approximated with the exchange 'metal working factory', this was chosen as metal is the main material constituting the final product. The general comment for the activity 'metal working faculty, GLO' [accessed 19.01.2018] specifies: '4.58E-10 units of this process are needed to produce 1 kg of final product. Additionally, the dataset for 'metal working factory' needs to be used together with the 'energy and auxilliary inputs, metal working factory', 1 kg of this exchange is required per 1 kg of final product. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
epoxy resin, liquid	0.138 kg
Comment: Data from Yuan et al. 2016. In terms of outputs assumed to have gone to waste PVC. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
epoxy resin, liquid	0.0291 kg
Comment: Data from Yuan et al. 2016. In terms of outputs assumed to have gone to waste PVC. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
flat glass, uncoated	1.49 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
flat glass, uncoated	0.307 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
heat, district or industrial, other than natural gas	19.3 MJ
Comment: Data from Yuan et al. 2016. Converted from kg of oil used for heating (0.46) to MJ of heat (19.26) using 'heat production, heavy fuel oil, at industrial furnace 1MW, GLO' [accessed 20 January 2018]. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
inert waste	-0.0561 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material, concrete block. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [2, 4, 1, 2, 2]	

inert waste	-0.144 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material, concrete block. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [2, 4, 1, 2, 2]	
iron scrap, unsorted	-0.92 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
kraft paper	0.337 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
kraft paper	0.963 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
metal working factory	3.43e-08 unit
Comment: Estimated value. The input of infrastructure is approximated with the exchange 'metal working factory', this was chosen as metal is the main material constituting the final product. The general comment for the activity 'metal working faculty, GLO' [accessed 19.01.2018] specifies: '4.58E-10 units of this process are needed to produce 1 kg of final product. Additionally, the dataset for 'metal working factory' needs to be used together with the 'energy and auxilliary inputs, metal working factory', 1 kg of this exchange is required per 1 kg of final product. Uncertainty distribution: lognormal; GSD2: 1.74; Pedigree matrix: [1, 4, 3, 2, 2]	
nylon 6	0.0331 kg
Comment: Data from Yuan et al. 2016. In terms of outputs assumed to have gone to waste PE. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
nylon 6	0.0669 kg
Comment: Data from Yuan et al. 2016. In terms of outputs assumed to have gone to waste PE. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
polyethylene, low density, granulate	0.2 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
polypropylene, granulate	7.47 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
polystyrene, general purpose	0.5 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
polyvinyl chloride, unspecified polymerisation, weighted average	0.1 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	

polyvinyl chloride, unspecified polymerisation, weighted average	0.93 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
printed paper	0.15 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
seal, natural rubber based	0.01 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
steam, in chemical industry	1.16 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
steam, in chemical industry	0.205 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
steel, low-alloyed, hot rolled	31.1 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
synthetic rubber	0.88 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
tap water	11.5 kg
Comment: Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
trisodium phosphate	0.52 kg
Comment: Used herein place of phosphating agent, degreaser and surface conditioning agent. In terms of outputs assumed to have gone to wastewater. Data from Yuan et al. 2016. Uncertainty distribution: lognormal; GSD2: 1.07; Pedigree matrix: [1, 4, 3, 2, 2]	
used cable	-0.63 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste aluminium	-4.35 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste aluminium	-1.34 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs.	

Uncertainty distribution: lognormal; **GSD2:** 1.06; **Pedigree matrix:** [1, 4, 1, 2, 2]

waste aluminium	-0.063 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste copper	-0.574 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste copper	-0.0269 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste copper	-1.21 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste glass	-5.46e-05 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste glass	-5.1e-06 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste glass	-0.000593 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste glass	-7.74e-06 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	

waste glass	-1.24e-06 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste glass	-0.00453 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste glass	-1.79 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste glass	-2.28e-05 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste packaging paper	-1.45 kg
Comment: Used here in place of kraft liner and paper. Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste plastic, mixture	-0.264 kg
Comment: Data from Yuan et al. 2016. Represents here waste ABS. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste plastic, mixture	-20.7 kg
Comment: Data from Yuan et al. 2016. Represents here waste ABS. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste plastic, mixture	-0.00729 kg
Comment: Data from Yuan et al. 2016. Represents here waste ABS. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste plastic, mixture	-0.285 kg
Comment: Data from Yuan et al. 2016. Represents here waste ABS.	

Production volume comment: Calculated from production volume of reference product using the relative outputs.

Uncertainty distribution: lognormal; **GSD2:** 1.06; **Pedigree matrix:** [1, 4, 1, 2, 2]

waste plastic, mixture	-0.0177 kg
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Comment: Data from Yuan et al. 2016. Represents here waste ABS.

Production volume comment: Calculated from production volume of reference product using the relative outputs.

Uncertainty distribution: lognormal; **GSD2:** 1.06; **Pedigree matrix:** [1, 4, 1, 2, 2]

waste plastic, mixture	-0.00693 kg
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Comment: Data from Yuan et al. 2016. Represents here waste ABS.

Production volume comment: Calculated from production volume of reference product using the relative outputs.

Uncertainty distribution: lognormal; **GSD2:** 1.06; **Pedigree matrix:** [1, 4, 1, 2, 2]

waste plastic, mixture	-0.00134 kg
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Comment: Data from Yuan et al. 2016. Represents here waste ABS.

Production volume comment: Calculated from production volume of reference product using the relative outputs.

Uncertainty distribution: lognormal; **GSD2:** 1.06; **Pedigree matrix:** [1, 4, 1, 2, 2]

waste plastic, mixture	-0.00295 kg
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Comment: Data from Yuan et al. 2016. Represents here waste ABS.

Production volume comment: Calculated from production volume of reference product using the relative outputs.

Uncertainty distribution: lognormal; **GSD2:** 1.06; **Pedigree matrix:** [1, 4, 1, 2, 2]

waste polyethylene	-3.66e-06 kg
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Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Also includes waste nylon.

Production volume comment: Calculated from production volume of reference product using the relative outputs.

Uncertainty distribution: lognormal; **GSD2:** 1.06; **Pedigree matrix:** [1, 4, 1, 2, 2]

waste polyethylene	-2e-05 kg
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Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Also includes waste nylon.

Production volume comment: Calculated from production volume of reference product using the relative outputs.

Uncertainty distribution: lognormal; **GSD2:** 1.06; **Pedigree matrix:** [1, 4, 1, 2, 2]

waste polyethylene	-1.9e-05 kg
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Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Also includes waste nylon.

Production volume comment: Calculated from production volume of reference product using the relative outputs.

Uncertainty distribution: lognormal; **GSD2:** 1.06; **Pedigree matrix:** [1, 4, 1, 2, 2]

waste polyethylene	-8.08e-06 kg
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Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Also includes waste nylon.

Production volume comment: Calculated from production volume of reference product using the relative outputs.

Uncertainty distribution: lognormal; **GSD2:** 1.06; **Pedigree matrix:** [1, 4, 1, 2, 2]

waste polyethylene	-4.84e-05 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Also includes waste nylon. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polyethylene	-0.000871 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Also includes waste nylon. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polyethylene	-0.000725 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Also includes waste nylon. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polyethylene	-0.298 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Also includes waste nylon. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polypropylene	-0.0311 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polypropylene	-7.4 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polypropylene	-0.000157 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polypropylene	-0.000857 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	

waste polypropylene	-0.0327 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polypropylene	-0.000814 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polypropylene	-0.00208 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polypropylene	-0.000346 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polystyrene	-0.00106 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polystyrene	-1.18e-05 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polystyrene	-5.37e-06 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polystyrene	-2.78e-05 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	

waste polystyrene	-0.00133 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polystyrene	-0.497 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polystyrene	-7.1e-05 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polystyrene	-2.93e-05 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polyvinylchloride	-0.000331 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polyvinylchloride	-0.012 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polyvinylchloride	-0.000134 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polyvinylchloride	-0.000314 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	

waste polyvinylchloride	-0.012 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polyvinylchloride	-6.05e-05 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polyvinylchloride	-1.17 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste polyvinylchloride	-0.0008 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste rubber, unspecified	-0.561 kg
Comment: This includes scrap synthetic, natural rubber (seal) and epoxy resin. Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste rubber, unspecified	-0.0132 kg
Comment: This includes scrap synthetic, natural rubber (seal) and epoxy resin. Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste rubber, unspecified	-0.316 kg
Comment: This includes scrap synthetic, natural rubber (seal) and epoxy resin. Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste steel	-10 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	

waste steel	-0.0127 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
waste steel	-21.1 kg
Comment: Data from Yuan et al. 2016. The amount is set equal to the input amount for the respective material. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
wastewater, average	-0.000126 m3
Comment: Converted from kg (1000 kg of water = 1 m3 of water). The amount is set equal to the input amount for the respective material. Data from Yuan et al. 2016. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
wastewater, average	-7.49e-05 m3
Comment: Converted from kg (1000 kg of water = 1 m3 of water). The amount is set equal to the input amount for the respective material. Data from Yuan et al. 2016. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
wastewater, average	-0.00627 m3
Comment: Converted from kg (1000 kg of water = 1 m3 of water). The amount is set equal to the input amount for the respective material. Data from Yuan et al. 2016. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	
wastewater, average	-0.00503 m3
Comment: Converted from kg (1000 kg of water = 1 m3 of water). The amount is set equal to the input amount for the respective material. Data from Yuan et al. 2016. Production volume comment: Calculated from production volume of reference product using the relative outputs. Uncertainty distribution: lognormal; GSD2: 1.06; Pedigree matrix: [1, 4, 1, 2, 2]	

Selected Impact Assessment Results

Method	Category	Indicator	Score
Ecological Scarcity 2021	total	UBP	1.12e+6 UBP
EF v3.1	material resources: metals/minerals	abiotic depletion potential (ADP): elements (ultimate reserves)	0.0176 kg Sb-Eq
IPCC 2021	climate change: total (excl. biogenic CO2)	global warming potential (GWP100)	3.84e+2 kg CO2-Eq

Direct Impact Contributions

Ecological Scarcity 2021, total, UBP: **1.12e+6 UBP**

Exchange	Amount	Impact (UBP)	Impact %
market for waste plastic, mixture RoW - waste plastic, mixture	-20.7 kg	1.96e+5	17.42%
market group for electricity, high voltage GLO - electricity, high voltage	1.34e+2 kWh	1.57e+5	13.95%
market for steel, low-alloyed, hot rolled GLO - steel, low-alloyed, hot rolled	31.1 kg	1.56e+5	13.87%
market for acrylonitrile-butadiene-styrene copolymer GLO - acrylonitrile-butadiene-styrene copolymer	21.2 kg	1.38e+5	12.3%
market for copper, cathode GLO - copper, cathode	1.81 kg	1.22e+5	10.86%
market for waste polypropylene RoW - waste polypropylene	-7.4 kg	7.13e+4	6.35%
market for aluminium, cast alloy GLO - aluminium, cast alloy	5.76 kg	6.68e+4	5.95%
market for waste polyvinylchloride RoW - waste polyvinylchloride	-1.17 kg	5.51e+4	4.91%
market for polypropylene, granulate GLO - polypropylene, granulate	7.47 kg	3.92e+4	3.49%
market for cable, unspecified GLO - cable, unspecified	0.63 kg	3.17e+4	2.82%
market for metal working factory GLO - metal working factory	3.43e-08 unit	1.62e+4	1.44%
market for kraft paper RoW - kraft paper	0.963 kg	8.93e+3	0.8%
market for trisodium phosphate GLO - trisodium phosphate	0.52 kg	7.98e+3	0.71%
market for waste polystyrene RoW - waste polystyrene	-0.497 kg	5.86e+3	0.52%
market for synthetic rubber GLO - synthetic rubber	0.88 kg	5.22e+3	0.47%

89 minor direct contributors have been omitted for clarity

EF v3.1, material resources: metals/minerals, abiotic depletion potential (ADP): elements (ultimate reserves): 0.0176 kg Sb-Eq

Exchange	Amount	Impact (kg Sb-Eq)	Impact %
market for copper, cathode GLO - copper, cathode	1.81 kg	0.013	73.78%
market for cable, unspecified GLO - cable, unspecified	0.63 kg	0.0031	17.63%
market for steel, low-alloyed, hot rolled GLO - steel, low-alloyed, hot rolled	31.1 kg	0.000495	2.82%
market for aluminium, cast alloy GLO - aluminium, cast alloy	5.76 kg	0.000449	2.55%
market for polypropylene, granulate GLO - polypropylene, granulate	7.47 kg	0.000111	0.63%
market for metal working factory GLO - metal working factory	3.43e-08 unit	8.66e-05	0.49%
market for acrylonitrile-butadiene-styrene copolymer GLO - acrylonitrile-butadiene-styrene copolymer	21.2 kg	7.31e-05	0.42%
market for compressed air, 1000 kPa gauge RoW - compressed air, 1000 kPa gauge	2.63 m3	6.46e-05	0.37%
market for trisodium phosphate GLO - trisodium phosphate	0.52 kg	4.34e-05	0.25%
market for synthetic rubber GLO - synthetic rubber	0.88 kg	3.88e-05	0.22%
market group for electricity, high voltage GLO - electricity, high voltage	1.34e+2 kWh	3.7e-05	0.21%
market for polyvinyl chloride, unspecified polymerisation, weighted average RoW - polyvinyl chloride, unspecified polymerisation, weighted average	0.93 kg	2.99e-05	0.17%
market for compressed air, 1000 kPa gauge RER - compressed air, 1000 kPa gauge	1.07 m3	2.64e-05	0.15%

market for flat glass, uncoated

RoW - flat glass, uncoated	1.49 kg	1.56e-05	0.09%
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market for epoxy resin, liquid

RoW - epoxy resin, liquid	0.138 kg	8.47e-06	0.05%
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89 minor direct contributors have been omitted for clarity

IPCC 2021, climate change: total (excl. biogenic CO₂), global warming potential (GWP100): **3.84e+2 kg CO₂-Eq**

Exchange	Amount	Impact (kg CO ₂ -Eq)	Impact %
market for acrylonitrile-butadiene-styrene copolymer GLO - acrylonitrile-butadiene-styrene copolymer	21.2 kg	98.9	25.77%
market group for electricity, high voltage GLO - electricity, high voltage	1.34e+2 kWh	90	23.45%
market for steel, low-alloyed, hot rolled GLO - steel, low-alloyed, hot rolled	31.1 kg	71.3	18.59%
market for aluminium, cast alloy GLO - aluminium, cast alloy	5.76 kg	36.6	9.53%
market for polypropylene, granulate GLO - polypropylene, granulate	7.47 kg	22.8	5.94%
market for waste plastic, mixture RoW - waste plastic, mixture	-20.7 kg	12.7	3.31%
market for copper, cathode GLO - copper, cathode	1.81 kg	12.2	3.18%
market for waste polypropylene RoW - waste polypropylene	-7.4 kg	5.04	1.31%
market for metal working factory GLO - metal working factory	3.43e-08 unit	4.67	1.22%
market for cable, unspecified GLO - cable, unspecified	0.63 kg	3.83	1.0%
market for synthetic rubber GLO - synthetic rubber	0.88 kg	2.79	0.73%
market for polyvinyl chloride, unspecified polymerisation, weighted average RoW - polyvinyl chloride, unspecified polymerisation, weighted average	0.93 kg	2.56	0.67%
market group for heat, district or industrial, other than natural gas GLO - heat, district or industrial, other than natural gas	19.3 MJ	2.02	0.53%

market for polystyrene, general purpose

GLO - polystyrene, general purpose	0.5 kg	1.88	0.49%
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market for waste rubber, unspecified

RoW - waste rubber, unspecified	-0.561 kg	1.78	0.46%
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89 minor direct contributors have been omitted for clarity

Source

First author	Yuan, Z.
Additional author(s)	Zhang, Y., and Liu, X.
Title	Life cycle assessment of horizontal-axis washing machines in China
Year	2016
Journal	International Journal of Life Cycle Asse
Volume number	21

First author	Michel, A.
Additional author(s)	Josephy, B., Bush, E., et al.
Title	Monitoring the washing machines market in Europe
Year	2015

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